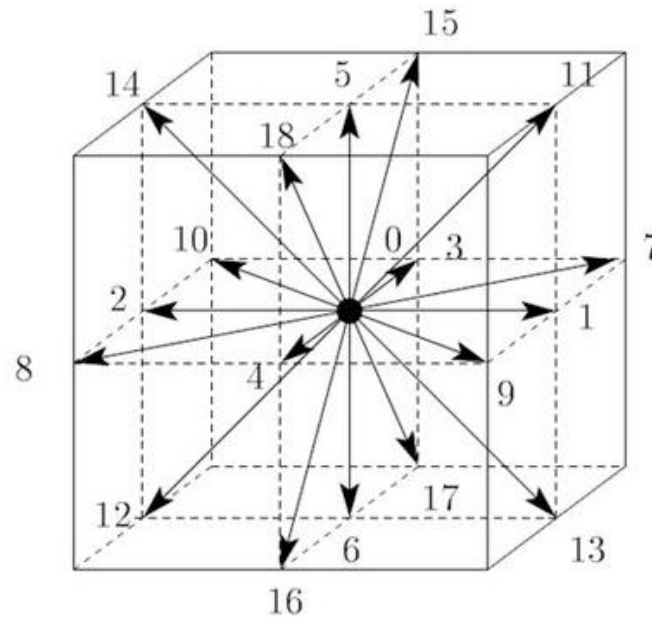




Lattice Boltzmann Method

Dr. Alessandro De Rosi

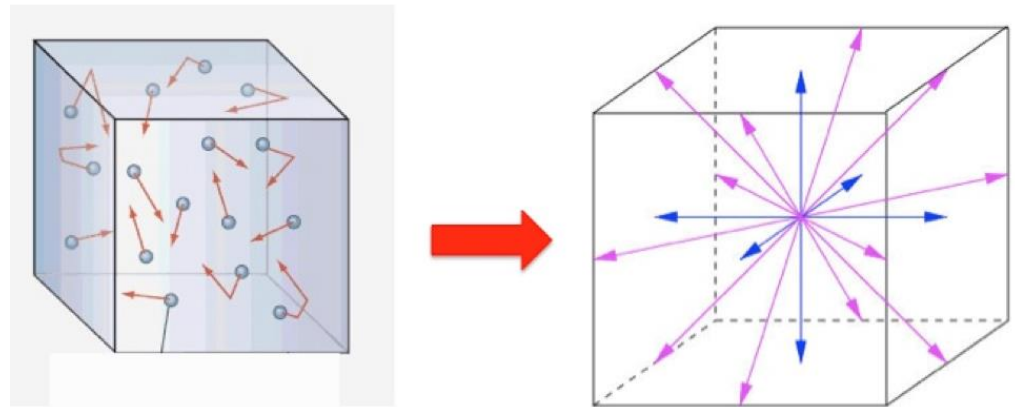
Slides adapted from Dr. Alistair Revell and Dr. Joseph O'Connor

Objectives of this Topic

- Provide a **theoretical basis of mesoscale approaches**
- Appreciate the **origins of the lattice Boltzmann method**
- Introduction to **kinetic theory & statistical mechanics**
- Understand the origin of the **equilibrium function**
- Have knowledge of the **discretisation of the LBM**
- Understand the **basic algorithm** of an LBM solver
- Understand basic **boundary conditions**
- Have knowledge of **further applications of the LBM**

Part 1: Overview

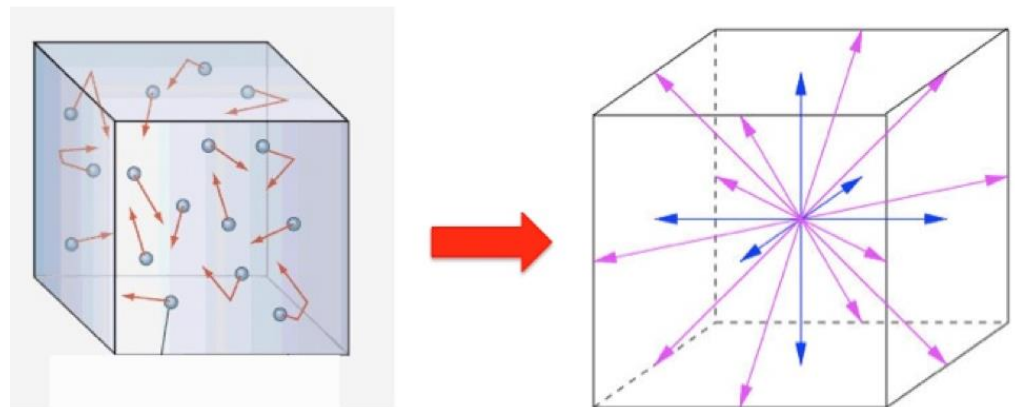
- Mesoscale dynamics
- Background to the lattice Boltzmann method
- Introduction to kinetic theory & statistical mechanics
- The Maxwell-Boltzmann distribution function



http://www.scholarpedia.org/article/Lattice_Boltzmann_Method

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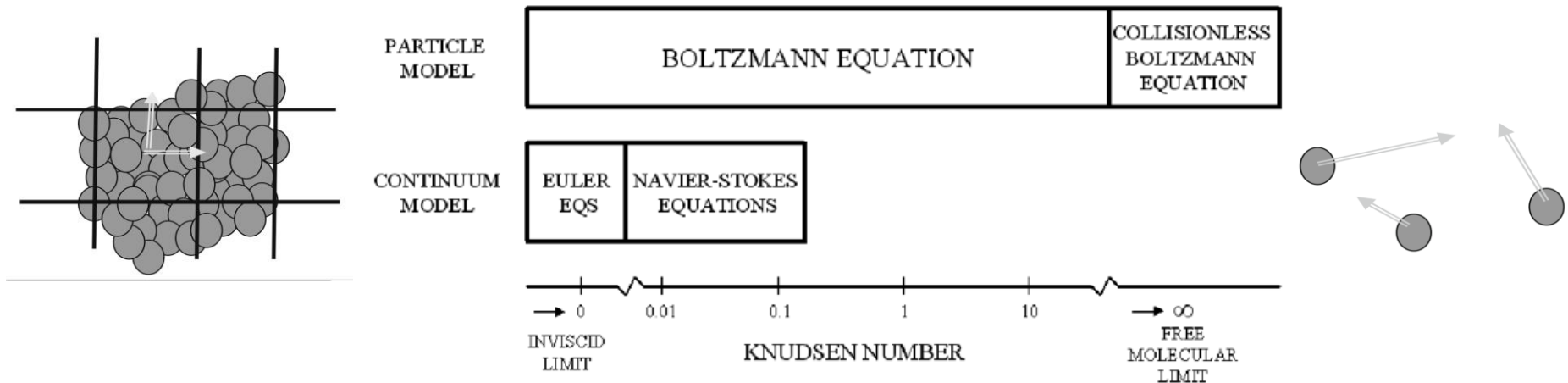


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Scales of Separation

■ Knudsen number

- Ratio between molecular free path and physical length scale



■ Two principles of mass and momentum transport

- Microscale: limit where $Kn \rightarrow \infty \rightarrow$ Molecular Dynamics
- Macroscale: limit where $Kn \rightarrow 0 \rightarrow$ Continuum (Navier Stokes)

Continuum Dynamics

- **Continuum (limit as $Kn \rightarrow 0$)**

- ODEs / PDEs obtained by applying conservation laws (mass, momentum, energy) to a (infinitesimally) small control volume

- **Governing equation is not trivial to solve**

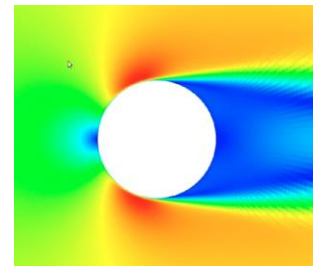
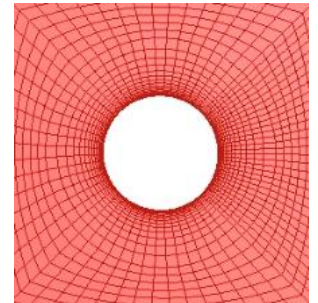
- Nonlinear, complex boundary conditions

- **Finite difference/volume/element schemes are used to convert the differential equations into a system of algebraic equations**

- Algebraic equations are applied at predefined spatial locations according to a mesh

- **Domain is decomposed into elements (mesh)**

- In theory each mesh represents a huge number particles (e.g. 10^{16})
 - Velocity, temperature, pressure of **all these particles** are represented by an average nodal value
 - The phenomenological quantities (viscosity, thermal conductivity, specific heat) are in general known by observation



Molecular Dynamics

- **For Molecular Dynamics, the medium is considered to be composed of many small particles**

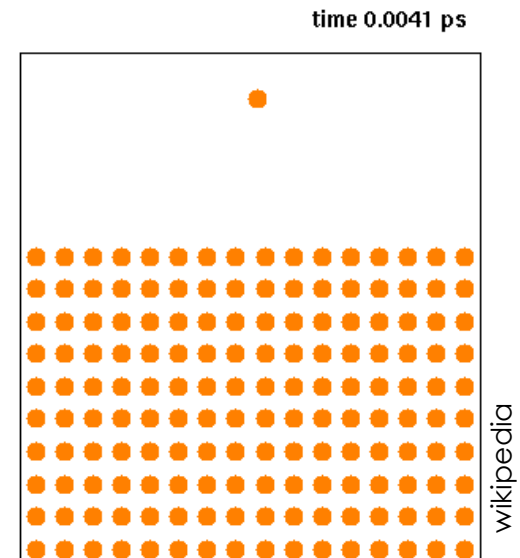
- Molecules travel in straight lines over finite distances before colliding with each other

- **Atoms/molecules are considered separately**

- Need to know inter-particle/molecular forces
- Must solve conservation of momentum for collisions
 - Therefore we need to know the position and velocity of each particle at each time step!

- **At this level there is no definition of temperature, pressure or thermo-physical properties such as viscosity, thermal conductivity etc.**

- Temperature and pressure are instead related to the kinetic energy of the particles, and frequency of the particle bombardment on the boundaries respectively



Molecular Dynamics

- **Physics in MD is relatively simple**

- Can handle phase change and complex geometries without any difficulty
- It is important to specify appropriate inter-particle force function

- **How many molecules in 18g water?**

- 18g is approximately 1 mole
- Avogadro's number: 6×10^{23}
- Number of molecules = number of moles x Avogadro's number

- **We need the position (x, y, z) and velocity (cx, cy, cz) for each particle**

- Timestep should be less than particle collision time
- On the order of Femto seconds (10^{-15})

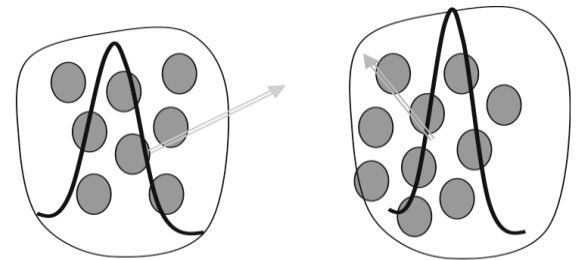
- **Main drawback of MD is computer resource and data reduction process**

- 18g of water in single precision would be approximately 10 yottabytes (10 trillion terrabytes)
- Even simulations on nm scale are impossible for MD

Mesoscale: Middle Ground

- Statistical mechanics is needed to translate from micro to macro
- In this room there are billions of molecules travelling at speeds of order 400m/s, like rockets hitting us!
 - Do we feel them?
 - Do we need to know the behaviour of each molecule?
- The motion of each individual particle is not important
 - It is the resultant effect of many particles that is important

- Main idea of the lattice Boltzmann method is to bridge the gap between micro and macro scale by not considering each particle behaviour alone, but a collection of particles as a unit
- The properties of a collection of particles is represented by a distribution function



Lattice Boltzmann Method

Continuum methods (macroscale)

- Based on the Navier Stokes equations
- Conservation of mass/momentum/energy on infinitesimal volume
- Finite (Volume/Element/Diff.) & Smooth Particle Hydrodynamics

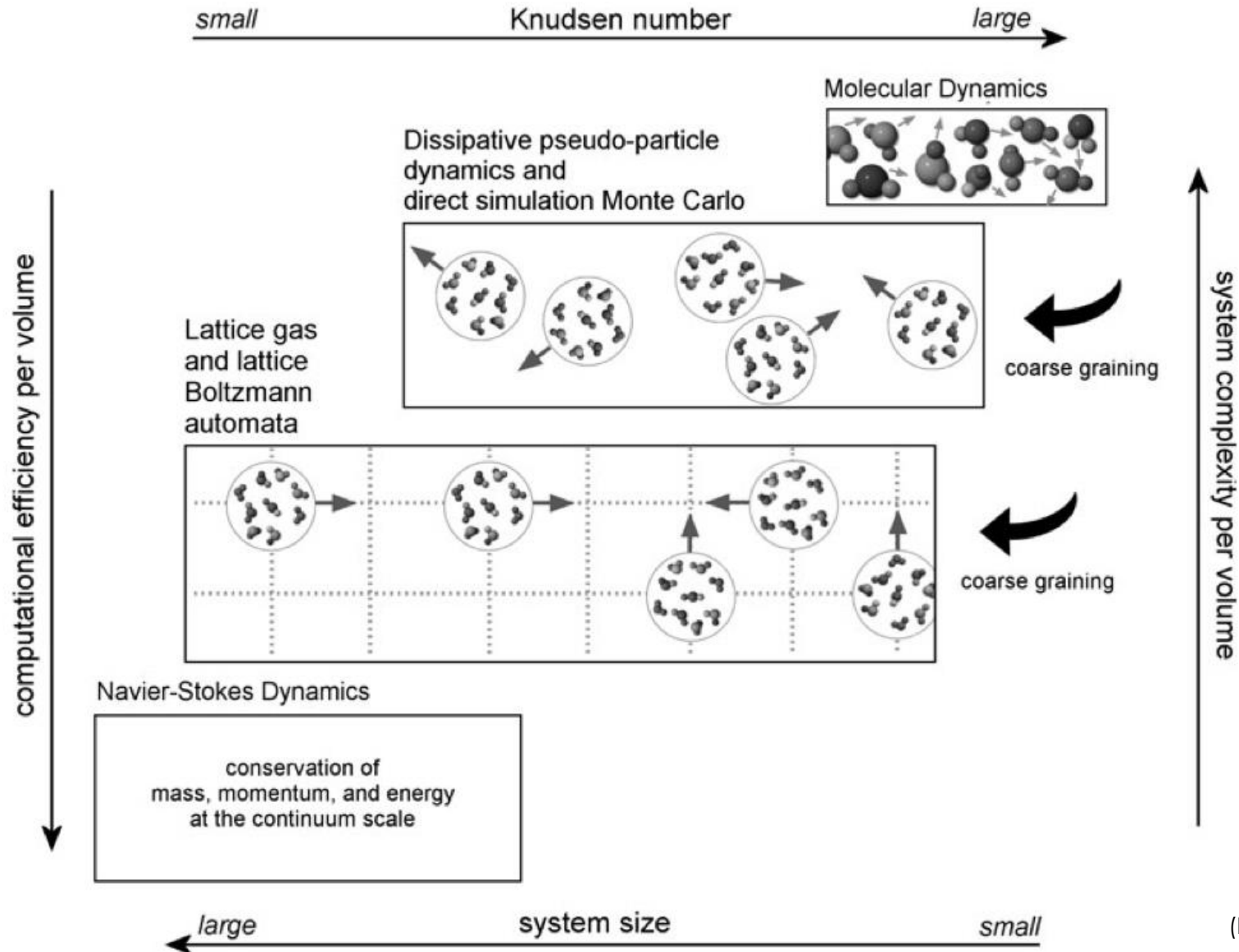
Mesoscale

- Based on kinetic theory: fits somewhere in the middle
- LBM falls within this category
- Instead of a single particle we consider a distribution function
- Distribution function represents a collection of particles

Molecular Dynamics (microscale)

- Small particles that collide with each other
- Inter-particle forces govern interactions
- At each timestep we must find trajectory of each particle
- Statistical mechanics is needed to relate to macroscale

From Micro to Macro



Discussion

- What applications might benefit from a **macroscopic** approach?
- What applications might benefit from a **microscopic** approach?
- What about a **mesoscopic** approach?

Commercial Activity

■ Two main players

- EXA PowerFlow: <http://www.exa.com/powerflow.html>
- NextLimit Xflow: <http://www.xflow-cfd.com/>

■ Their videos are extremely impressive

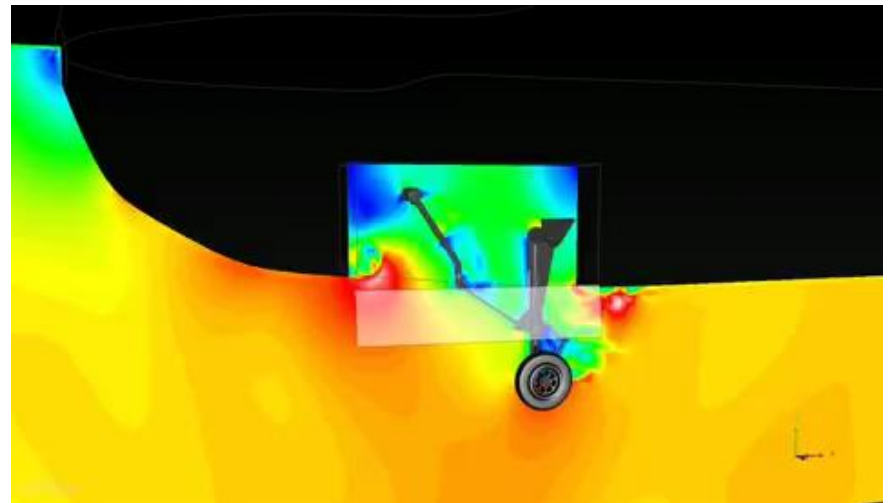
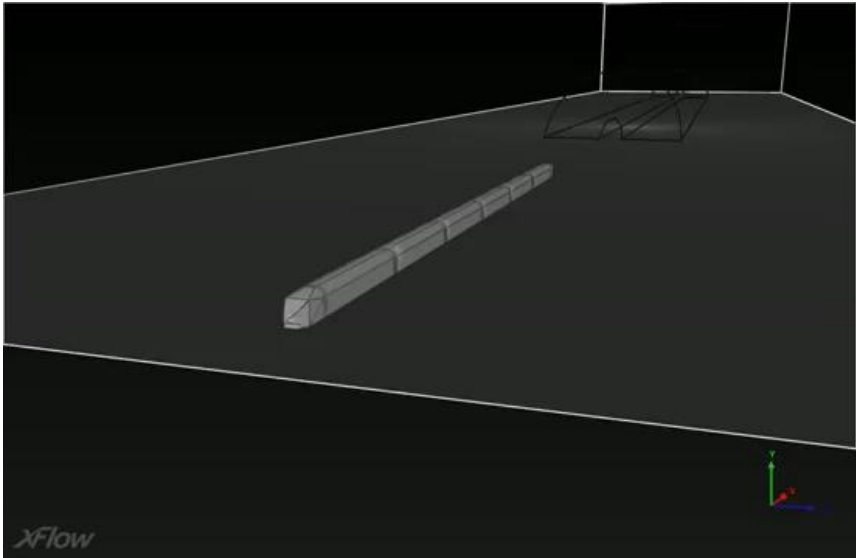
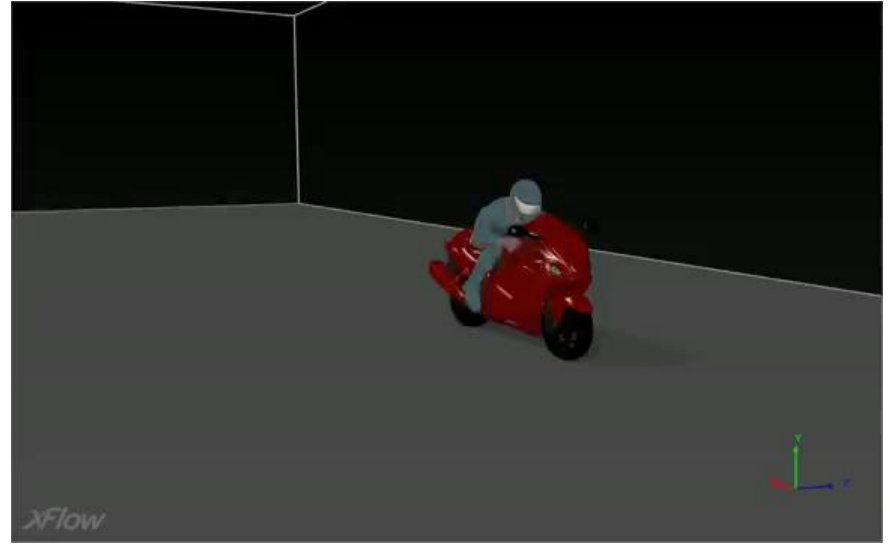
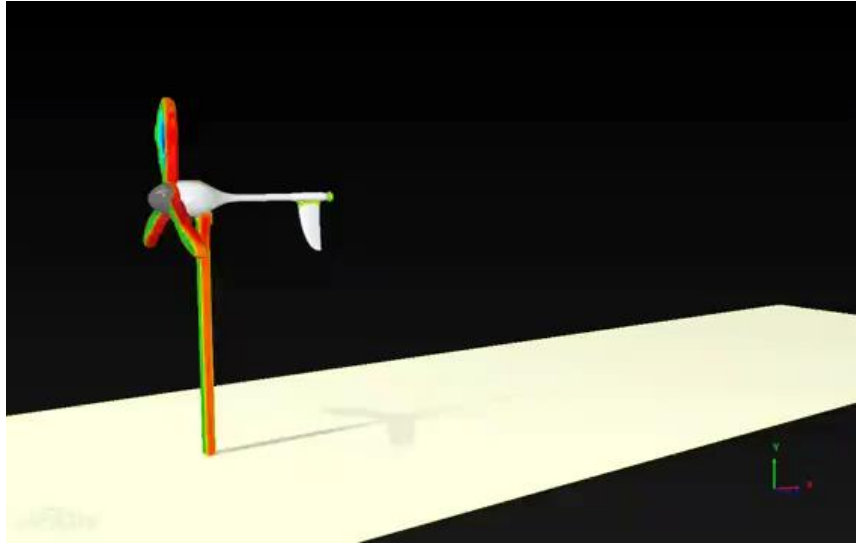
- But this is not so surprising...NextLimit started by selling special effects to film/gaming industry
- Then someone asked the question: maybe they can be made accurate?

■ Growing very fast

- PowerFLOW 15+ years
- XFLOW deal with MSc (NASTRAN)
- Airbus, JLR, Renault have been using LBM for some time now

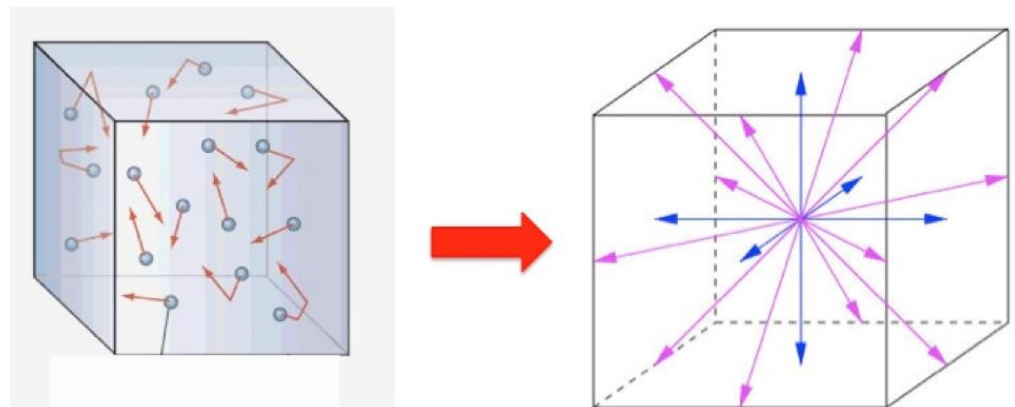


Commercial Activity



Part 1: Overview

- Mesoscale dynamics
- Background to the lattice Boltzmann method
- Introduction to kinetic theory & statistical mechanics
- The Maxwell-Boltzmann distribution function



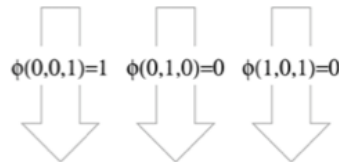
http://www.scholarpedia.org/article/Lattice_Boltzmann_Method

Origins of the LBM

- **19th Century: Boltzmann & Maxwell**
- **1940: Cellular Automata: John von Neumann**
 - Self replicating systems
- **1976: Hardy, Pomeau and de Pazzis (HPP)**
- **1986: Lattice Gas Cellular Automata (FHP)**
 - Noisy
- **1989: Lattice Boltzmann Equations**
 - Eliminated statistical noise
- **1992: Use of BGK model in lattice Boltzmann equation**
 - More efficient and allowed relations between transport coefficients (e.g. viscosity)
- **1997: Derived from kinetic theory**
 - A complete theoretical framework
 - Demonstrates that LBM approaches Navier Stokes equations
 - LBM extended even beyond Navier Stokes equations

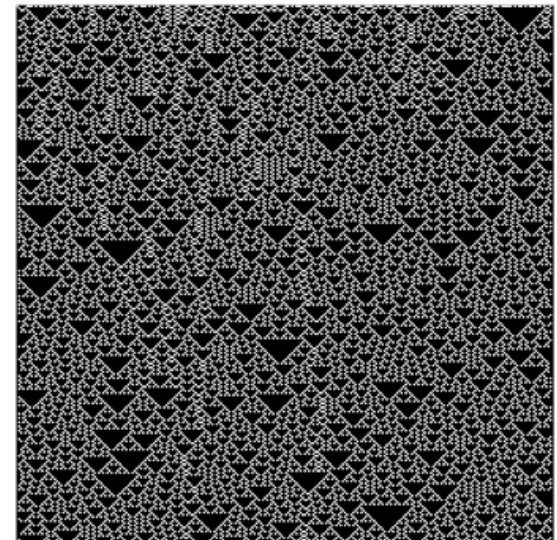
Cellular Automata

- An algorithm that occupies a position in space and interacts with its identical neighbours
 - Invented in the 1940s by von Neumann to model a self-replicating system
- A cellular automaton **examines** its own state and the states of its neighbours at any particular timestep and then **resets** its own state for the next timestep according to simple rules
 - Hence, the rules and the initial and boundary conditions imposed on the group of cellular automata uniquely determine their evolution in time



Sukop

$$\begin{aligned}
 \phi(0, 0, 0) &= 0 \\
 \phi(0, 0, 1) &= 1 \\
 \phi(0, 1, 0) &= 0 \\
 \phi(0, 1, 1) &= 0 \\
 \phi(1, 0, 0) &= 1 \\
 \phi(1, 0, 1) &= 0 \\
 \phi(1, 1, 0) &= 0 \\
 \phi(1, 1, 1) &= 0
 \end{aligned}$$



Wolfram (1986)

Conway's Game of Life

- Devised by British mathematician John Conway (inspired by von Neumann)
 - Introduced in Scientific American (1970)
 - Popularised Cellular Automata
- 'Zero player' game: entirely determined from initial state and rules
 - A live cell with < 2 live neighbours *dies, by under-population*
 - A live cell with 2 or 3 live neighbours *lives on to the next generation*
 - A live cell with > 3 live neighbours *dies, by overcrowding*
 - A dead cell with 3 live neighbours *becomes a live cell, by reproduction*
- Principle that complex behaviour results from a simple set of rules
 - Analogous to dynamical systems

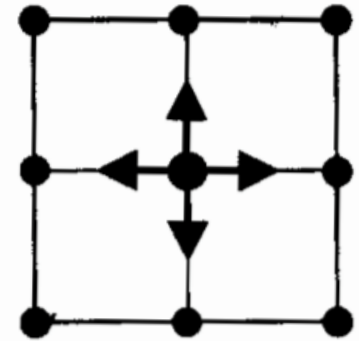


(wikipedia!)

Lattice Gas Cellular Automata

■ 1976: Hardy, Pomeau and de Pazzis (HPP)

- Restrict particle motion to four directions (Left, Right, Up, Down)
- All particles have the same mass
- Nearly works for fluids but still not physical



■ Fluid dynamics is represented by Boolean operations

- Particles can move only along one of the four directions defined by the discrete vector displacements indicated by the arrows.
- In a timestep, particles hop to the nearest neighbour pointed by the corresponding discrete vector; longer and shorter jumps are forbidden
- This means all particles have the same energy
- No two particles sitting on the same site can move in the same direction (exclusion principle)

■ Rules are applied to ensure that mass and momentum are conserved

- Two mechanisms of motion: **Collide** and **Stream**
- All pre and post-collision momenta must live on the lattice

LGCA - HPP Model

- It is not rotationally invariant
 - Consider the stress tensor in 2D case

$$T_{ab} \equiv \rho u_a u_b + P \delta_{ab}.$$

$$T_{xx} = P + \rho u^2,$$

$$T_{xy} = T_{yx} = \rho uv,$$

$$T_{yy} = P + \rho v^2,$$

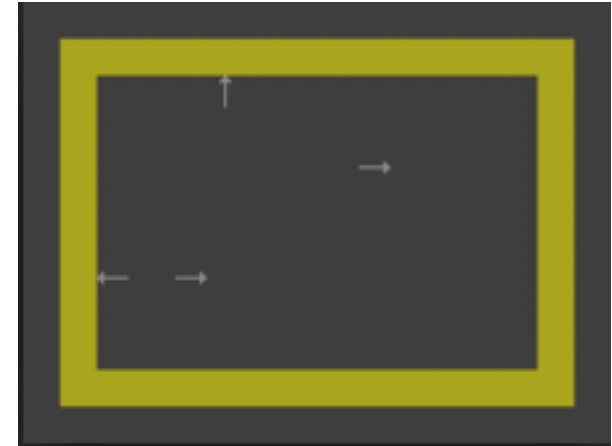
- In HPP model this becomes

$$T_{xx} = P + \rho(u^2 - v^2),$$

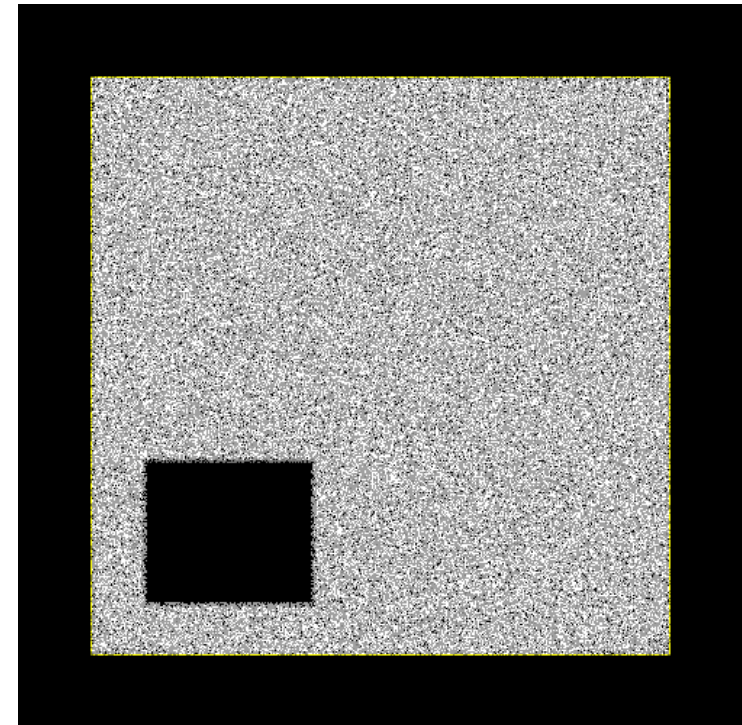
$$T_{xy} = T_{yx} = 0,$$

$$T_{yy} = P + \rho(v^2 - u^2).$$

- Does not capture shear stress
 - Well known to give square vortices!
-
- Cannot be solved by grid refinement
 - It is a mathematical limitation of the approach

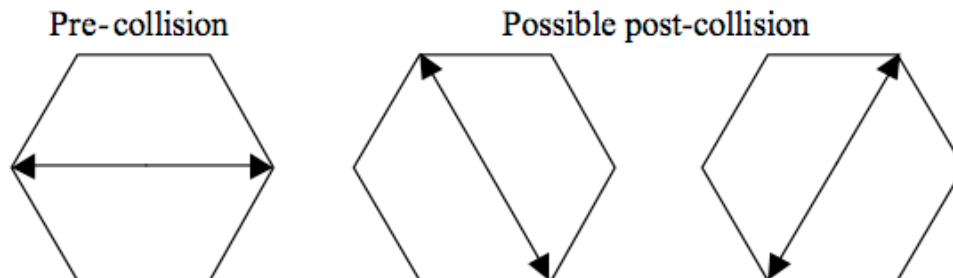
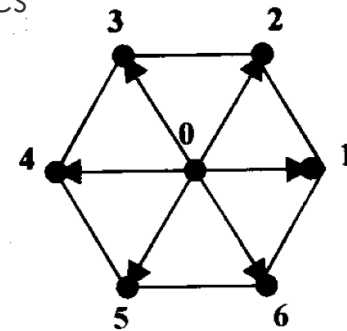


(wikipedia)



LGCA – FHP Model

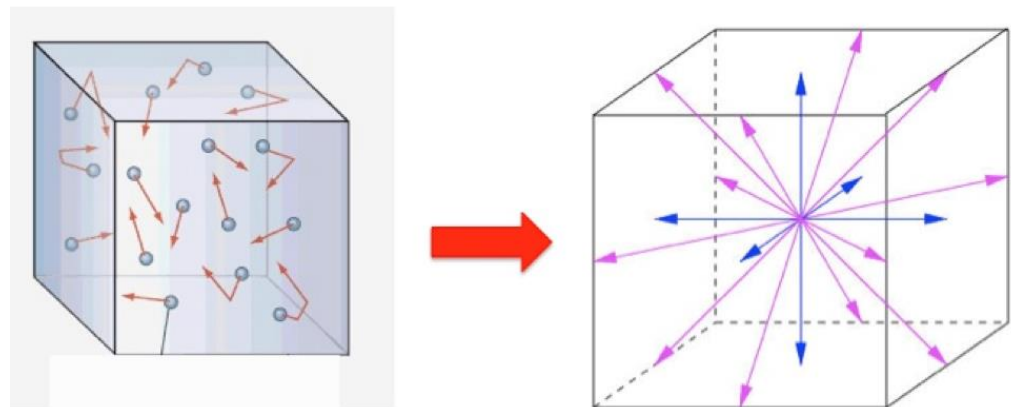
- **1986: Frisch, Hasslacher and Pomeau (FHP)**
 - A simple cellular automaton obeying nothing but conservation laws at a microscopic scale was able to reproduce complexity of real fluid dynamics
 - Great excitement...made the New York Times
- **Extra directions enough to ensure realistic fluid dynamics**
- **Important random element introduced**
- **Used look up tables to find collision outcome**



	Bit Value							
	128 R	64 S	32 F	16 E	8 D	4 C	2 B	1 A
0	0	0	0	0	0	0	0	0
1	0	0	0	0	0	0	0	1
2	0	0	0	0	0	0	1	0
3	0	0	0	0	0	0	1	1
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255	1	1	1	1	1	1	1	1

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Kinetic Theory

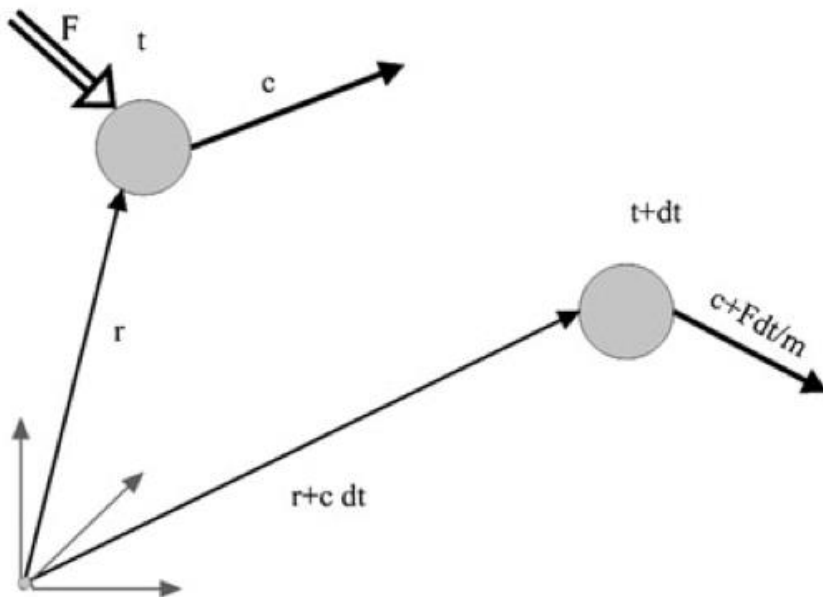
- Complete set of position ($\mathbf{x}$) and momentum ($\mathbf{p}$) coordinates for all individual particles gives exact dynamical state of system
- Together with classical mechanics, allows exact prediction of future states
 - However, this level of description is essentially not possible

Statistical Mechanics

- Represent system by ensemble of many copies
- Use distribution function $\mathbf{f}$ to denote probability of finding given conditions (position and speed)
 - $6N$ variables
 - Changes in $\mathbf{f}$ with time given by Liouville/BBGKY theorem

Kinetic Theory

- **Main building block of materials in nature is molecules and sub-molecules**
 - Particle and molecule are interchangeable terms in our considerations
- **Consider solid spheres moving randomly in conservative fashion in free space**
 - Satisfies conservation of mass, momentum and energy
 - Particle has velocity $\mathbf{c}$



- Rate of change of momentum is equal to applied force:

$$\mathbf{F} = \frac{d(m\mathbf{c})}{dt}$$

- For constant mass:

$$\mathbf{F} = m \frac{d\mathbf{c}}{dt} = m\mathbf{a}$$

- Position is found from velocity:

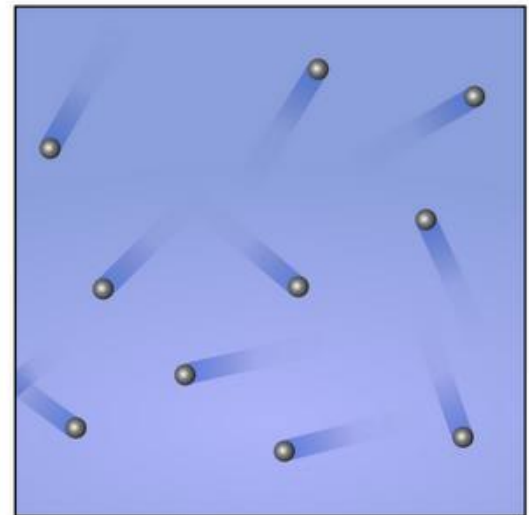
$$\mathbf{c} = \frac{d\mathbf{r}}{dt}$$

- Velocity becomes:

$$\mathbf{c} + \mathbf{F} dt / m$$

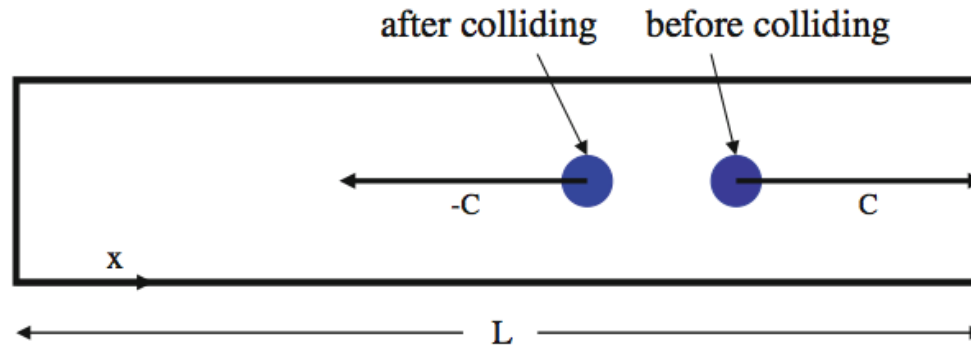
Kinetic Theory: Particle View

- As magnitude of velocity increases, the internal energy of the system is increasing
- This increase in KE of the particle is referred to macroscopically as an increase in the temperature
- Particles continuously bombard container walls (or other objects)
- The force exerted per unit area is referred to as pressure in macroscopic terms



Kinetic Theory: Example

- Single molecule in container length L cross section area A :



- Change of momentum at end of container:

$$\Delta p_x = p_{1,x} - p_{2,x} = mc_x - (-mc_x) = 2mc_x$$

- Time between the particle hitting the right side wall:

$$\Delta t = \frac{2L}{c_x}$$

- Single molecule x force:

$$F_x = \frac{\Delta p_x}{\Delta t} = \frac{mc_x^2}{L}$$

Kinetic theory: Pressure

- Generalising for N particles, force in x direction is:

$$F_x = \frac{Nm\overline{c_x^2}}{L}$$

- Thus total force under isotropic ($c_x = c_y = c_z$) conditions:

$$\overline{c^2} = \overline{c_x^2} + \overline{c_y^2} + \overline{c_z^2} \longrightarrow F = \frac{Nm\overline{c^2}}{3L}$$

- Pressure (force/unit area):

$$P = \frac{F}{L^2} = \frac{Nm\overline{c^2}}{3V}$$

- Note the link between pressure and kinetic energy:

$$PV = \frac{2}{3}\text{KE}$$

- This neglects molecular interaction and molecule sizes however this gives a surprisingly good approximation at room temperature

Kinetic theory: Temperature

- Experimentally we know that the ideal gas equation is:

$$PV = MRT = Nk_B T$$

- Where $M = N/N_A$ is the number of moles (N_A is Avogadro's number)
- R is the gas constant
- The Boltzmann constant is $k_B = R/N_A$

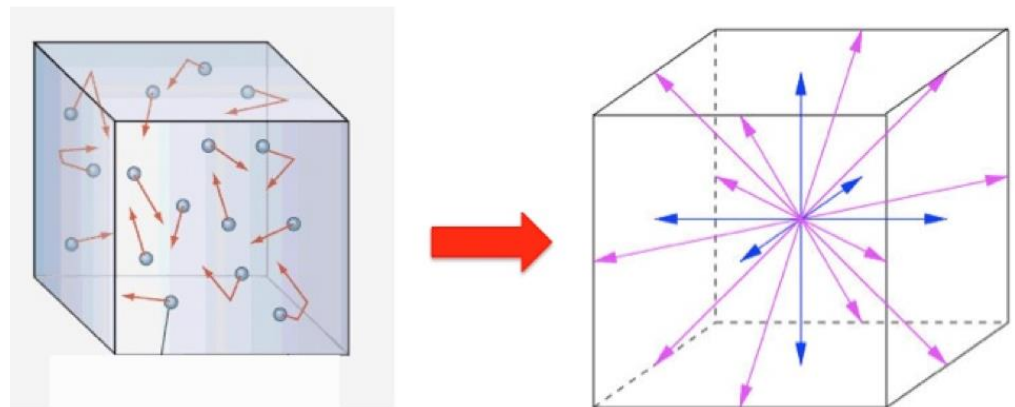
- Thus temperature is:

$$T = \frac{2}{3} \frac{\text{KE}}{Nk_B}$$

- Both temperature and pressure in macroscopic world are simple scalings of kinetic energy in the microscopic world

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Distribution Function



- **Maxwell (1831-1879): Scottish theoretical physicist**
- **Recognised that dealing with a huge number of particles was impractical (even though governing equations were known exactly)**
 - Representative values can be used instead
 - For example, analogy of a class representative vs 200 students
- **Key step is realising that in most cases knowledge of the velocity and position of each particle at each instance of time is not necessary**
 - In a gas there is a wide range of particle velocities, what is the representative value?
- **Proposed a distribution function to characterise the effect of particles**
 - In other words, what percentage of the particles in a certain location at a given instance in time have velocities within a certain range
 - The distribution function is not a function of time

Maxwell-Boltzmann Equilibrium

- In a gas that is left undisturbed, the particle collisions will tend to evenly distribute the particle velocities about their mean velocity
 - This state can be described by the Maxwell-Boltzmann equilibrium distribution
 - The equilibrium distribution is a function of the relative particle velocity ($\mathbf{v} = \mathbf{c} - \mathbf{u}$)

- We assume the 3D distribution is separable in each dimension:

$$f^{eq}(v) = f_x^{eq}(v_x) f_y^{eq}(v_y) f_z^{eq}(v_z)$$

- With $v = \text{constant}$ we have:

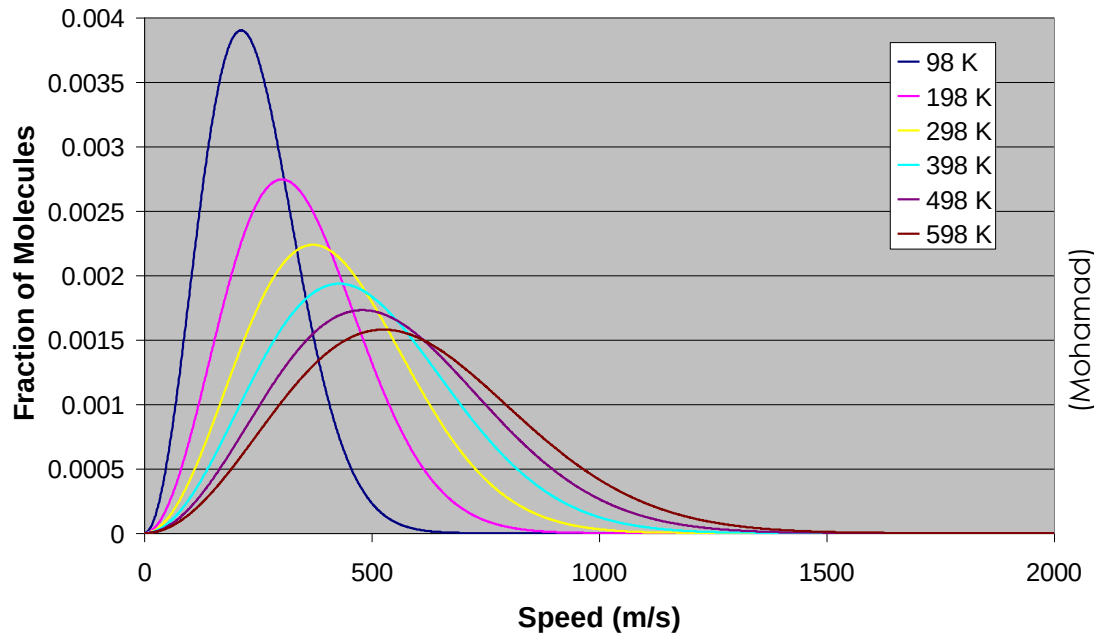
$$\ln(f_x^{eq}(v_x)) + \ln(f_y^{eq}(v_y)) + \ln(f_z^{eq}(v_z)) = \text{const}$$

- Which is true if:

$$f_x^{eq}(v_x) = e^a e^{-bv_x^2} \longrightarrow f^{eq}(v) = e^{3a} e^{-b(v_x^2 + v_y^2 + v_z^2)} = e^{3a} e^{-bv^2}$$

Maxwell-Boltzmann Distribution

- Expanding the moments for **a** and **b** gives: $f^{eq}(v) = \rho \left(\frac{m}{2\pi k_B T} \right)^{3/2} e^{-\frac{mv^2}{2k_B T}}$



- Increases parabolically from zero for low speeds, reaches a maximum and then decreases exponentially
- As temperature increases, position of maximum shifts to right
- Total area under curve is always one

Maxwell-Boltzmann Distribution

- For air (nitrogen) at 0 and 100 degrees Celsius
- Most probable speed:
 - Find maximum of the function
- Average speed:
 - Integration of f from zero to infinity

$$\langle v \rangle = \int_0^{\infty} v f^{eq}(v) dv$$

- Root mean average speed:

$$v_{rms} = \sqrt{\langle v^2 \rangle} = \left(\int_0^{\infty} v^2 f^{eq}(v) dv \right)^{1/2} = \sqrt{\frac{3k_B T}{m}}$$

